

# eROSITA × DESI × AION-flow — Experiment Pipeline

Living document — keep in sync with the actual pipeline. Last updated: 2026-07-17. Render to PDF with python docs/render\_pipeline\_pdf.py (writes docs/pipeline.pdf).

## 1. Goal

Use a frozen astronomical foundation model (AION-1 base) as an information probe: predict eROSITA X-ray properties (broad-band flux, luminosity, hardness) and host stellar mass from non-X-ray DESI modalities (optical spectra, WISE, redshift, Legacy imaging) with a probabilistic normalizing-flow head. Report likelihood gain over a prior, not just point  $R^2$ . This repo extends the PAI26 paper with (a) a cleaned crossmatch and (b) uncertainty-aware, per-target training.

## 2. Compute

- FASRC** (Harvard SLURM). Account `finkbeiner_lab` (confirmed working 2026-07-21); smokes on `gpu_test`, real runs on `gpu`. `siag_gpu` is *not* visible to us yet — it needs `siag_lab` group membership (Coldfront request pending).
- Env: `module load Miniforge3/26.1.0-fasrc01 + conda env + cu124 torch wheel + pip install -e . polymathic-aion zuko safetensors wandb`. Built at `$AIONFLOW_ROOT/envs/aionflow` (torch 2.6.0+cu124).
- Workflow**: develop locally → `git push` → `git pull` on the cluster; the SSH ControlMaster socket `~/ssh/cm-fasrc` is used only to submit jobs and move data. All real work runs in `sbatch` jobs on compute nodes (login-node compute gets killed by arbiter2).
- Tracking: Weights & Biases** (`--wandb`, project `erosita-desi-aion-flow`). Logs config, per-step batch NLL, per-epoch train/val NLL, and the final per-combination test metrics table. Auto-falls back to offline if a compute node lacks outbound internet (`wandb sync` later); a tracking failure never kills a job.

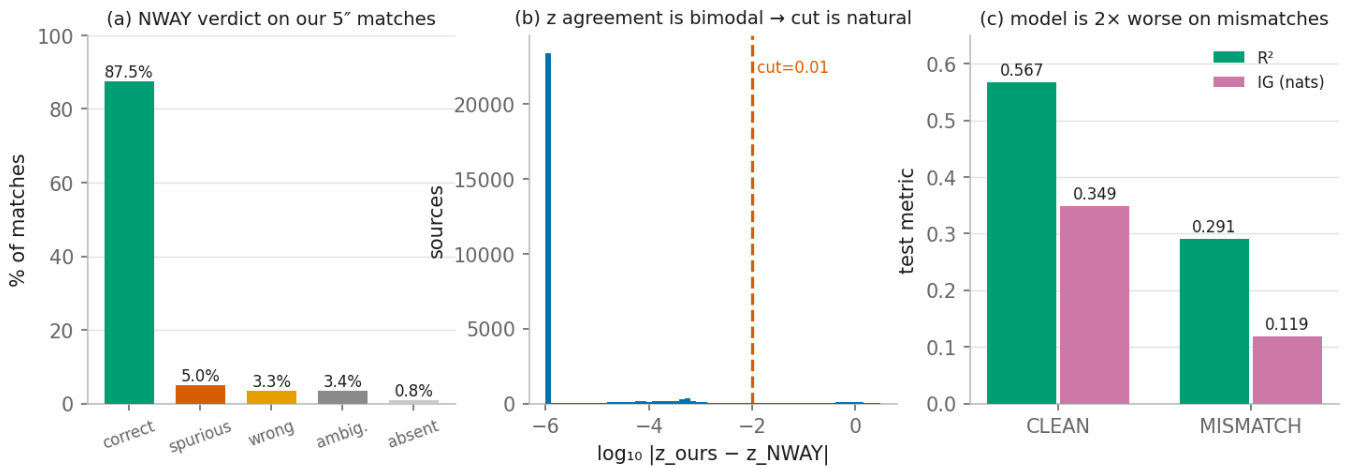
## 3. Data sources & download

| Modality                | Source                                                     | Notes                                                                                                                                                                     |
|-------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| X-ray catalog           | SRG/eROSITA-DE DR1 / eRASS1 (Merloni+2024)                 | broad band <code>ML_FLUX_1</code> + sub-bands <code>P1–P4</code> rates/fluxes <b>with asymmetric errors</b> (LOWERR/UPERR). Delivered as the eROSITA×DESI match CSVs.     |
| Optical spectra         | DESI DR1 coadd spectra                                     | per-healpix <code>coadd-{survey}-{program}-{healpix}.fits</code> , trimmed to matched <code>targetids</code> , compiled to <code>erosita_spectra_merged_32k.hdf5</code> . |
| Imaging                 | Legacy Survey DR10 griz cutouts                            | 160×160 px <code>fits_pool/&lt;targetid&gt;.fits</code> via <code>legacysurvey.org</code> cutout service (~11 GB).                                                        |
| Redshift / galaxy props | DESI DR1 VAC                                               | <code>z</code> , <code>logmstar</code> (photometric), emission lines, <code>spectype</code> (QSO/GALAXY/STAR).                                                            |
| Foundation model        | <code>polymathic-ai/aion-base</code> (HuggingFace, ~95 MB) | frozen tokenizer + backbone.                                                                                                                                              |

All inputs are local; only `aion-base` is internet-bound. "Adding sources our match missed" would require new DESI DR1 spectrum downloads (deferred).

## 4. Crossmatch & cleaning

- **Original match (existing):** naive 5" nearest-neighbor (`crossmatch_radec.py`, `astropy`) between DESI target RA/Dec and the eROSITA catalog → 33,938 pairs → 30,441 unique DESI targets → **28,646 with a retrieved spectrum**.
- **Quality problems found:** separations pile up near the 5" cut (median 3.08", 52% >3"); **14% of eROSITA sources match >1 DESI target** (26% of pairs in multi-match groups) → the same X-ray label duplicated onto several spectra.
- **Cleaning (NWAY):** validate against **Salvato et al. 2025 (A&A 704, A344)** — the eRASS1 Bayesian NWAY counterpart catalog — via VizieR J/A+A/704/A344/dr1ls10 (join on IAUName; compare their counterpart's zsp to our matched DESI z). Per-targetid class: **87.5% correct, 4.6% spurious, 3.3% wrong, 3.4% ambiguous, 0.8% absent** → keep = correct filter (`data/match_quality.csv`).
- **Sanity checks (cleaning is sound, 3 independent angles):** (a)  $|z_{\text{ours}} - z_{\text{NWAY}}|$  is **bimodal** — 88.7% <1e-4 (same object), 3.0% >0.1 (different), only 0.6% in the 0.01–0.1 valley → the z-cut is natural, not tuned; (b) mismatches show larger match separation (3.19" vs 3.06") but only weakly — separation can't discriminate at eROSITA's ~arcsec position errors, which is *why* prior-based NWAY is needed; (c) the trained model is **2× worse on mismatches** ( $R^2$  0.29 vs 0.57) — independent confirmation the flags localize real failures.
- **Pipeline order (IMPORTANT):** clean → **dedup to the NWAY counterpart per targetid** → **re-derive the train/val/test split on the cleaned sample** (targetid-grouped 80/10/10, seed 42, leakage-safe) → stage. Splits must be redone *after* cleaning.



## 5. Targets & uncertainties

Four separate 1-D flow runs (`data/targets_extra.csv` carries the derived targets + errors, keyed by `targetid`):

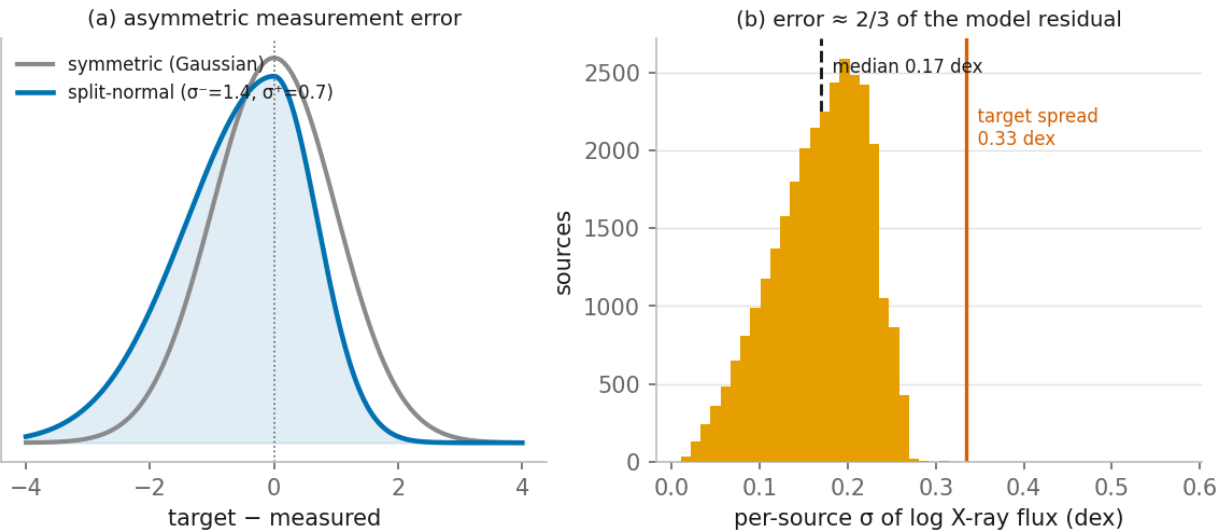
| Target                     | Source                                | Error model                                                                                                                                                                                                                                               |
|----------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>log_ml_flux_1</code> | <code>log10(ML_FLUX_1)</code>         | <b>split-normal</b> (two-piece Gaussian) from LOWERR/UPERR:<br>$\text{sig\_hi} = \log_{10}(1 + \text{UPERR}/F)$ ,<br>$\text{sig\_lo} = -\log_{10}(1 - \text{LOWERR}/F)$<br>(capped). Median $\approx 0.17$ dex; <b>98% asymmetric, low side heavier</b> . |
| <code>log_lx</code>        | <code>ML_FLUX_1 + z</code> (Planck18) | same dex $\sigma$ as flux.                                                                                                                                                                                                                                |
| <code>logmstar</code>      | DESI VAC photometric mass             | <b>spectype systematic floor</b> : 0.20 dex (GALAXY) / 0.30 dex (QSO). No per-source posterior $\sigma$ available (PROVABGS covers only 0.4% of this AGN sample). Caveat: 82% of sample is QSO → mass AGN-contaminated; evaluate split by spectype.       |

|                 |                              |                                                                                                                                                                                                                                     |
|-----------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hr32 (hardness) | $(P3-P2)/(P3+P2)$ band rates | modeled in <b>arctanh space</b><br>$u = \text{arctanh}(HR)$ ,<br>$\text{sig}_u = \text{sig}_{hr} / (1 - HR^2)$ . Only ~17% have $S/N > 2$ ( $\sigma \approx \text{signal}$ ) → <b>IG-primary</b> , $R^2$ secondary. HR43/T dropped. |
|-----------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Targets & error treatment (four separate V1 runs, all --error-mode convolve)

| target        | source                       | error model                     | mode     | trainable                  |
|---------------|------------------------------|---------------------------------|----------|----------------------------|
| log_ml_flux_1 | eROSITA ML_FLUX_1            | split-normal (LO/HI), ~0.17 dex | convolve | full sample ✓              |
| log_lx        | flux + z (Planck18)          | same dex $\sigma$ as flux       | convolve | full (z-dominated)         |
| logmstar      | DESI photometric mass        | spectype floor 0.2/0.3 dex      | convolve | clean for GALAXY; 82% QSO  |
| hr32_u        | eROSITA band rates (arctanh) | propagated $\sigma_{HR}$        | convolve | IG-primary; ~17% $S/N > 2$ |

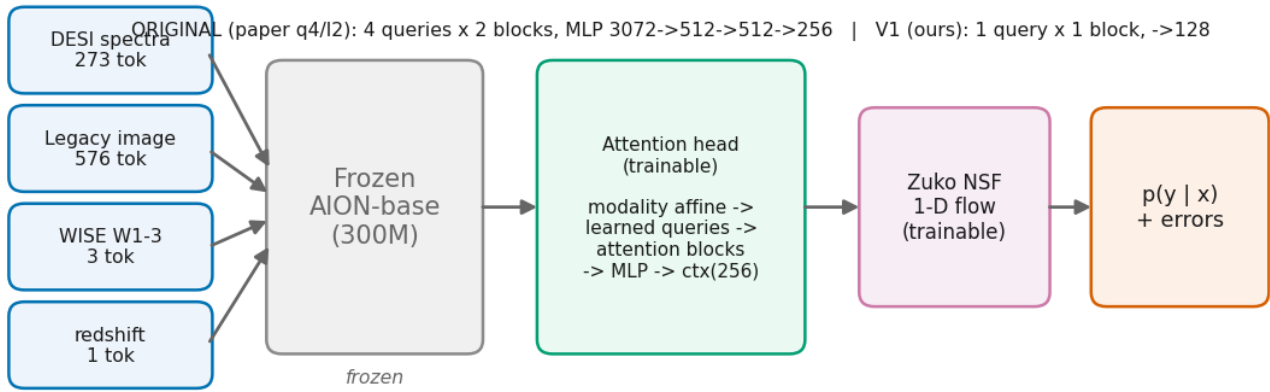
**How errors enter the flow (split-normal convolution likelihood):**  $NLL_i = -\log \int p_{\text{flow}}(t \mid x_i) \cdot K(y_i \mid t; \text{sig}_{\text{lo}_i}, \text{sig}_{\text{hi}_i}) \, dt$ , a fixed 1-D quadrature (~21–41 nodes over  $\pm 4\sigma$ ),  $\text{error\_mode} \in \{\text{none}, \text{inject}, \text{convolve}\}$  (default **convolve** = deconvolves; **inject** broadens; **none** = paper behavior).



Measurement error is large relative to signal: for flux,  $\sigma \approx 0.17$  dex is  $\sim 2/3$  of the model's residual variance → the model sits near the measurement-noise floor, so error-aware training + a reported error floor matter.

**Deferred error refinement:** X-ray sub-band non-detections are **upper limits (censored)**, not missing/large-error; proper handling via a censored likelihood (ref 2022A&A...661A...3S). First pass uses split-normal  $\sigma$  on measured values.

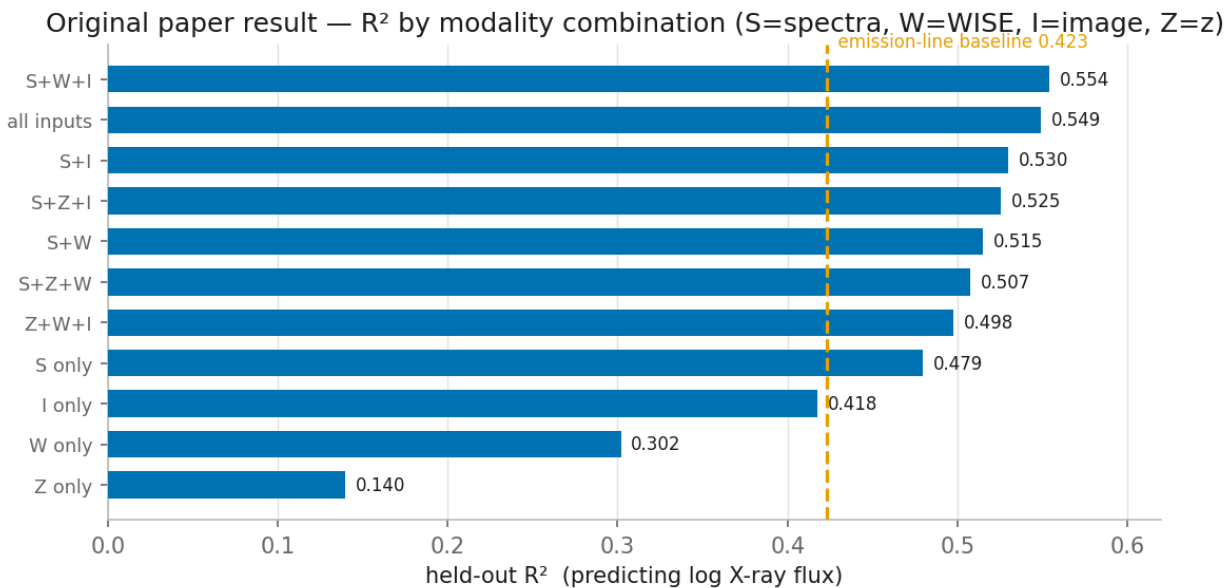
## 6. Model architecture



Frozen **AION-base** tokenizer + backbone → per-modality token sequences (spectra=273, image=576, WISE=3, z=1; missing modalities omitted). Trainable head + flow:

- **V1 minimal head** (AIONAttentionContext, parameterized): per-modality affine calibration → **1 learned query + 1 attention block (8 heads)** cross/self-attention → 768 → MLP (→128) → 256-d context. (~7.8M params vs the paper's 4-query/3072-d ~16.8M.)
- **Flow**: Zuko conditional **NSF**, 1-D target, 8 transforms, context 256; standardized target; **KDE (Scott)** prior as the IG reference.
- Per-batch a modality combo is sampled; AION frozen; AdamW; checkpoint best all-inputs val NLL.
- **V2 (later)**: replace frozen AION with a self-trained attention encoder over the raw spectrum.

**Original paper result** (q4/I2, epoch-13) —  $R^2$  by modality combination; full spectra beat the emission-line baseline, all-modality is best:



## 7. Training & evaluation

- Four separate V1 runs (--target {log\_ml\_flux\_1, log\_lx, logmstar, hr32}), --clean (NWAY keep), --error-mode convolve, on gpu; smoke on gpu\_test with a --limit subset.
- **Metrics**: primary **IG / exp(IG)** vs KDE prior (esp. HR); secondary  **$R^2$  + reduced- $\chi^2$**  using per-source  $\sigma$  + the measurement-error floor; **clean-vs-full delta**; per-spectype split for logmstar/HR.
- Every run is mirrored to **W&B** (--wandb), so training curves are watchable live and runs are comparable across targets/error-modes via the run tags (<target>, <error\_mode>, V1).

## 8. Key findings so far

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- Naive 5" match is **~8% wrong** per NWAY; the paper's q4/l2 model is **~2× worse on the mismatches ( $R^2$  0.29 vs 0.57)**; dropping them lifts test  $R^2$  from the published **0.549** → **0.567** (no retraining).
- X-ray targets are measurement-noise-dominated (flux  $\sigma \approx 0.17$  dex; HR  $\sigma \approx$  its own spread) → uncertainty handling is essential.

## 9. Status

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- ■ Repo cloned; V1 head coded+tested; FASRC scaffolding + **verify-before-push staging scripts** (scripts/fasrc\_preflight.sh, scripts/fasrc\_stage\_data.sh).
- ■ Sidecars built: data/match\_quality.csv (clean filter), data/targets\_extra.csv (targets + split-normal/arctanh errors).
- ■ **Error-aware flow** implemented + tested (split\_normal\_log\_kernel, convolve\_logprob, ConditionalNSFFlow.log\_prob\_convolved; 14/14 tests, recovers analytic Gaussian convolution).
- ■ **Clean** → **dedup** → **re-split** implemented + tested (build\_clean\_manifest; 25,200 clean rows, exact 80/10/10, leakage-safe, deterministic).
- ■ **Full wiring done + validated locally**: prepare-data --clean cleans/dedups/re-splits + writes extra target & error columns; AIONHDF5Dataset emits per-source  $\sigma$  (10-tuple); main.py --target {..., logmstar, hr32\_u} / --error-mode {none, inject, convolve} / --clean; batch\_nll uses log\_prob\_convolved. 14/14 tests; a real prepare-data --clean --limit 12 produced correct columns.
- ■ **FASRC ready**: 17 GB data staged + fits\_pool unzipped (10,355 cutouts); conda env built and verified (torch 2.6.0+cu124, aion, zuko all import); repo is a real git clone of multitarget-clean-errors; account finkbeiner\_lab confirmed via sbatch --test-only.
- ■ **W&B tracking** wired into train (opt-in --wandb, offline-safe, no-op if absent). First live run confirmed end-to-end on gpu\_test.
- ■ **Staged-data validation** (scripts/validate\_staged.py + 13 tests + sbatch/validate\_staged.sbatch), wired as a hard gate in train.sbatch: schema, row-count alignment, dedup/leakage/split-fractions, NWAY clean filter,  $\sigma$  positivity & coverage, value ranges, fits coverage, and the dataloader 10-tuple contract.
- ■■ **Caught by that validation**: the fits\_pool unzip had been interrupted (10,355 of 27,373 cutouts on disk). Because staging drops any object lacking a cutout, the first full run staged **9,592 rows instead of ~22k** with no error. Fixed (count-based unzip idempotency + a coverage check that fails above 20% loss); re-staging.
- ■ **One canonical staged copy** (staged\_paper, 30,762 rows = the paper layout incl. its 3,389 duplicate rows; test = 3,054 rows **exactly matching the published table's n\_test**) + the cleaned configuration as a **runtime view** (clean\_split.csv, 25,200 targetids; verified EXACT vs the materialized version, which was then deleted). Superset validation: 103 passed / 0 failed; its extreme values (log\_lx=48.07 at z=4.71, logmstar<5) are all NWAY-rejected wrong matches — independent confirmation the clean filter cuts exactly the garbage.
- ■ **Smokes** (plumbing only, 21 effective steps each): V1-clean-convolve  $R^2=0.29$ ; paper-head repro  $R^2 \approx 0.02$  (big head needs real training). Both end-to-end incl. wandb + 15-combo eval.
- ■ **Running now (order changed by decision: light-first, reproduction deferred)**: V1 head + cleaned view + --error-mode convolve, log\_ml\_flux\_1, 15 epochs, batch 448, A100-80 (gpu, job 34083239) — the first results-grade run of the new-science config.
- ■ Then: **paper-table reproduction** (full q4/l2 head, --error-mode none, native paper splits, 50 epochs) → compare to published 0.549/0.554; consider the AION **token cache** (~185 GB per-combo, ~5–15× epoch speedup) before the 4-target production sweep; siag\_gpu once siag\_lab membership is granted.
- ■■ siag\_lab membership pending (Coldfront) → siag\_gpu invisible for now; using gpu\_test (smokes) / gpu (real runs) under finkbeiner\_lab.
- Known refinement: hr32\_u piles at the arctanh clip for low-S/N HR (huge  $\sigma$ ) — fine for flux/lx/logmstar; the HR run wants an S/N gate/clip.

## 10. Deferred / out of scope

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SFR (needs DESI FastSpecFit VAC),  $\Gamma$  (=HR relabel), X-ray upper-limit censoring, full Salvato re-crossmatch of "the rest", V2 self-trained encoder, saliency/attribution.

